

Exercise 6C**Sketching Reciprocal Functions****FOUNDATION**

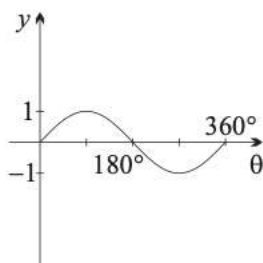
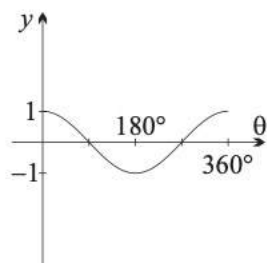
Note: Question **9** and **13** concern the graphs of the three trigonometric functions $\sin \theta$, $\cos \theta$, and $\tan \theta$. The graphs are drawn in the questions, but readers unfamiliar with them may like to delay attempting these questions until after studying Chapter 7.

- 1 a** Sketch $y = \frac{1}{x-1}$ after carrying out the following steps.
 - i** State the natural domain, and find the y-intercept.
 - ii** Find any points where $y = 1$ or $y = -1$.
 - iii** Explain why $y = 0$ is a horizontal asymptote.
 - iv** Draw up a table of values and examine the sign.
 - v** Identify any vertical asymptotes, and use the table of signs to write down its behaviour near any vertical asymptotes.
- b** Repeat for $y = \frac{2}{3-x}$.
- c** Repeat for $y = -\frac{2}{x+2}$.
- d** Repeat for $y = \frac{5}{2x+5}$.
- 2** Follow steps (i)–(v) of Question **1a** to investigate the function $y = \frac{2}{(x-1)^2}$ and hence sketch its graph. Show any points where $y = 1$.
- 3** Investigate the domain, zeroes, sign and asymptotes of the function $y = -\frac{1}{(x-2)^2}$ and hence sketch its graph. Show any points where $y = -1$.
- 4 a** Let $f(x) = x + 1$.
 - i** Graph $y = f(x)$ showing the intercepts with the axes.
 - ii** Also show the points where $f(x) = 1$ and $f(x) = -1$.
 - iii** Hence on the same number plane sketch $y = \frac{1}{f(x)}$.
- b** Follow similar steps for the function $g(x) = 2 - x$.

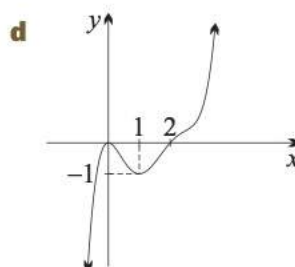
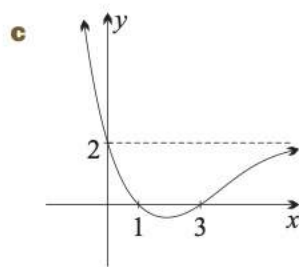
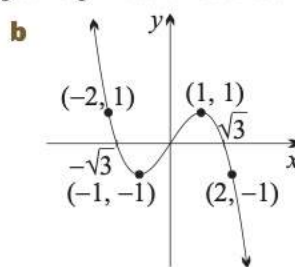
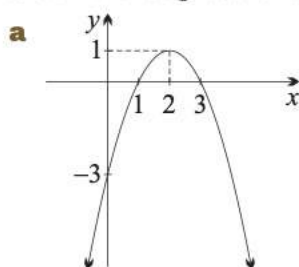
DEVELOPMENT

- 5** Let $y = f(x)$ where $f(x) = \frac{1}{3}(x+1)(x-3)$.
 - a** Show that $y = 1$ at $x = 1 - \sqrt{7}$ and $x = 1 + \sqrt{7}$. Plot these points.
 - b** Complete the graph of $y = f(x)$ showing the vertex, the intercepts with the axes and the points where $f(x) = -1$.
 - c** What is the range of $y = f(x)$?
 - d** Hence sketch $y = \frac{1}{f(x)}$ on the same number plane.
 - e** What is the range of $y = \frac{1}{f(x)}$?
- 6 a** Follow similar steps to Question **5** to sketch $y = \frac{1}{4}(4-x^2)$ and $y = \frac{4}{4-x^2}$.
 - b** What is the range of $y = \frac{1}{4}(4-x^2)$?
 - c** What is the range of $y = \frac{4}{4-x^2}$?
- 7 a** Sketch $y = \frac{1}{2}(x^2 + 1)$ showing the points where $y = 1$.
 - b** What is the minimum value of $\frac{1}{2}(x^2 + 1)$?
 - c** Sketch $y = \frac{2}{x^2 + 1}$ on the same number plane.

- d** Explain why the x -axis is an asymptote to $y = \frac{2}{x^2 + 1}$.
- e** What is the maximum value of $\frac{2}{x^2 + 1}$?
- 8** Let $f(x) = -x^2 + 2x - 3$.
- a** What is the maximum of $f(x)$?
- b** Sketch $y = f(x)$ and $y = \frac{1}{f(x)}$ on the same number plane.
- c** Explain why the x -axis is an asymptote for $y = \frac{1}{f(x)}$.
- d** What is the minimum of $\frac{1}{f(x)}$?
- 9** The diagrams show first $y = \cos \theta$ then $y = \sin \theta$ for $0^\circ \leq \theta \leq 360^\circ$.



- a i** Copy the sketch of $y = \cos \theta$, and add to it the sketch of $y = \frac{1}{\cos \theta}$.
- ii** What are the domain and range of $y = \frac{1}{\cos \theta}$ in this interval?
- b i** Copy the sketch of $y = \sin \theta$, and add to it the sketch of $y = \frac{1}{\sin \theta}$.
- ii** What are the domain and range of $y = \frac{1}{\sin \theta}$ in this interval?
- 10** Prove that the symmetry of a function is preserved when taking reciprocals. That is, prove that the reciprocal of an even function is an even function, and prove that the reciprocal of an odd function is an odd function.
- 11 a** Graph $y = \frac{2+x}{x}$ by first noting that $y = 1 + \frac{2}{x}$.
- b** Hence graph $y = \frac{x}{2+x}$.
- 12** Sketch the reciprocal of each function shown, showing all significant features.



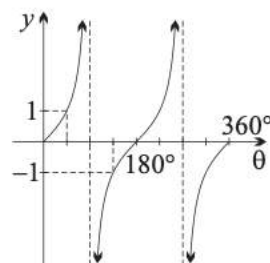
- 13** The sketch shows $y = \tan \theta$ for $0^\circ \leq \theta \leq 360^\circ$.

- a** What is the domain of $y = \tan \theta$ in this interval?
b Where is $\tan \theta = 0$ in this interval?
c Hence state the domain of $y = \frac{1}{\tan \theta}$ in this interval.

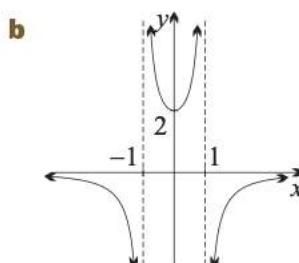
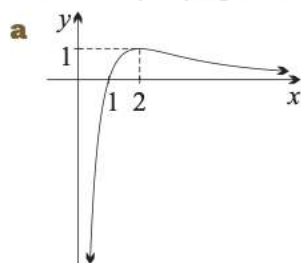
- d** What is $\lim_{\theta \rightarrow 90^\circ} \frac{1}{\tan \theta}$?

- e** Copy the graph of $y = \tan \theta$, and add to it the graph of $y = \frac{1}{\tan \theta}$.

- f** What is the range of the reciprocal function?



- 14** In each case the graph of $y = f(x)$ is given. Sketch the graph of $y = \frac{1}{f(x)}$, paying careful attention to the domain, any asymptotes, and any relevant limits.



ENRICHMENT

- 15** The arguments in the solution to Example 15a seems to rely on the following assertion about the graphs of a function $y = f(x)$ and its reciprocal function $y = (f(x))^{-1}$:

‘When one curve has a local maximum, the other has a local minimum.’

This statement is not strictly true. State the qualification that needs to be made in this statement, and give an example where the qualification is necessary.

- 16** Let $y = \frac{1}{x-2}$. Write down precisely the equation of the reciprocal function.

FOUNDATION

- c**
-

-

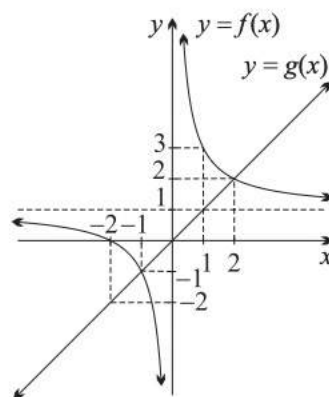
-

ii $y = f(x) - g(x)$

- 8** Graphs the functions $y = x^2$ and $y = |x|$ on the same set of axes.
- a i** Explain why the graph of $y = x^2 + |x|$ must lie above both original graphs.
- ii** Hence sketch the sum $y = x^2 + |x|$. Take care with the shape at $x = 0$.
- b i** Explain why the graph of $y = x^2 - |x|$ is on or below the x -axis for $-1 \leq x \leq 1$.
- ii** Hence sketch the difference $y = x^2 - |x|$. Take care with the shape at $x = 0$.
- 9 a** Sketch $y = \sqrt{x}$ and $y = x$, paying careful attention to the domain and to the points where these graphs intersect.
- b** Hence sketch $y = x - \sqrt{x}$.

- 10** The graph on the right is a repeat of the one used in the theory to investigate the sum of functions that involves an asymptote. In this question you will investigate their difference

$$d(x) = f(x) - g(x).$$



- a** The given curves intersect at $x = -1$ and $x = 2$. What feature of the graph of $y = d(x)$ occurs at these values?
- b** Write down the zero of $f(x)$ and hence evaluate $d(x)$ there.
- c** Evaluate $d(1)$ by using the ordinates of $f(x)$ and $g(x)$.
- d** Copy and complete each statement relating to the vertical asymptote of $y = f(x)$.
- i** As $x \rightarrow 0^+$, $f(x) \rightarrow \dots$ and $g(x) \rightarrow \dots$ so $d(x) \rightarrow \dots$
- ii** As $x \rightarrow 0^-$, $f(x) \rightarrow \dots$ and $g(x) \rightarrow \dots$ so $d(x) \rightarrow \dots$
- e** Copy and complete each statement relating to the horizontal asymptote of $y = f(x)$.
- i** As $x \rightarrow +\infty$, $f(x) \rightarrow \dots$ and $g(x) \rightarrow \dots$ so $d(x) \rightarrow \dots$
- ii** As $x \rightarrow -\infty$, $f(x) \rightarrow \dots$ and $g(x) \rightarrow \dots$ so $d(x) \rightarrow \dots$
- f** Briefly explain why $d(x)$ approaches $y = 1 - x$ for large values of x .
- g** Sketch $y = d(x)$ showing all these features.
- 11** For each pair of functions, first graph $y = f(x)$ and $y = g(x)$ on the same number plane. Then in a different colour graph the sum $y = f(x) + g(x)$.
- a** $f(x) = \frac{1}{x}$ and $g(x) = x$
- b** $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x+2}$
- c** $f(x) = 2^x$ and $g(x) = x - 1$
- d** $f(x) = 2^x$ and $g(x) = 2^{-x}$
- 12** For each pair of functions in the previous question, sketch the difference $y = f(x) - g(x)$.

- 13** It may be that if $y = f(x)$ and $y = g(x)$ exhibit even or odd symmetry then we know the symmetry of $s(x) = f(x) + g(x)$ and of $d(x) = f(x) - g(x)$.
- a** Try to complete the following table, predicting the symmetry of the resulting functions, and test your claims on the examples in this exercise.

	$f(x)$ even, $g(x)$ even	$f(x)$ odd, $g(x)$ odd	$f(x)$ even, $g(x)$ odd
$s(x)$			
$d(x)$			

- b** Prove formally that if $f(x)$ and $g(x)$ are both odd then $s(x)$ is also odd. That is, prove that $s(-x) = -s(x)$ for all x in its domain.

ENRICHMENT

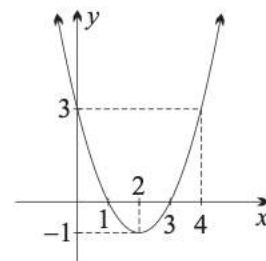
- 14** Consider the two functions $f(x) = \frac{1}{2}x - 3$ and $g(x) = 1 + \frac{1}{x-1}$. The purpose of this question is to sketch the graph $y = f(x) + g(x)$.
- a** Find the x -coordinates of the points where $f(x) = -g(x)$.
- b** Plot the line $y = f(x)$ and the hyperbola $y = g(x)$ on the same number plane, and mark the points found in part **a**.
- c** Show that $f(x) + g(x) - (\frac{1}{2}x - 2) \rightarrow 0$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. Add the line $y = \frac{1}{2}x - 3$ to your graph. This is an *oblique asymptote* to the curve $y = f(x) + g(x)$.
- d** Complete your sketch of the sum $y = f(x) + g(x)$.
- 15** In the text, the sum $s(x) = f(x) + g(x)$ was drawn for the functions $f(x) = \frac{2}{x} + 1$ and $g(x) = x$, and the difference $d(x) = f(x) - g(x)$ was drawn in Question **10**. The oblique asymptotes of $s(x)$ and $d(x)$ were also drawn. This question now finds the equations of the oblique asymptotes to the two curves by the method used in the previous question.
- a** Simplify $s(x) = f(x) + g(x)$, and show that $s(x) - (x + 1) = \frac{2}{x}$. Hence explain why $y = x + 1$ is an oblique asymptote to the curve $y = s(x)$.
- b** Similarly simplify $d(x) = f(x) - g(x)$, and so find its oblique asymptote.

Exercise 6E

Modifying a Function using Absolute Value

FOUNDATION

- 1** The graph of $y = f(x)$ is sketched to the right. To draw each transformation, copy the graph and draw the transformed graph in a different colour on the same axes.



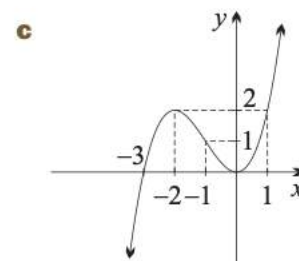
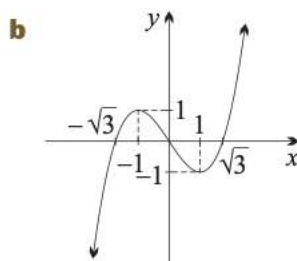
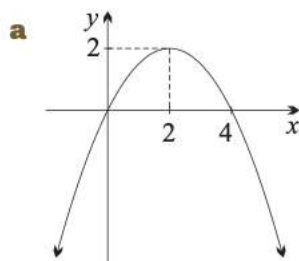
- a** Replace any part of the graph below the x -axis by its reflection above the x -axis. This will give you the graph of $y = |f(x)|$.
- b** Graph only the parts of $y = f(x)$ that are to the right of the y -axis. Then add the reflection of these parts in the y -axis. This will give you the graph of $y = f(|x|)$. Notice the symmetry in the y -axis.
- c** Using the graph of $y = f(|x|)$, replace any part of the graph below the x -axis by its reflection above the x -axis. This gives the graph of $y = |f(|x|)|$.
- d** Using the graph of $y = |f(x)|$, graph only the parts that are to the right of the y -axis. Then add the reflection of these parts in the y -axis. What do you notice about the result of this and the answer to part **c**?

- 2** In each case, follow the steps of Question **1** and use the given graph of $y = f(x)$ to sketch:

i $y = |f(x)|$

ii $y = f(|x|)$

iii $y = |f(|x|)|$



- 3** Sketch each given function and use it to then sketch:

i $y = |f(x)|$

ii $y = f(|x|)$

iii $y = |f(|x|)|$

a $y = x - 1$

b $y = 2^{-x} - 1$

c $y = \frac{1}{1-x}$

- 4 a** Explain why the graphs of $y = 2^x$ and $y = |2^x|$ are the same.
- b** Write $y = 2^{|x|}$ using cases, then sketch its graph.

DEVELOPMENT

- 5 a** Sketch $y = f(x)$ where $f(x) = (x+1)(x-2)$, showing all x -intercepts and the vertex.

b Hence sketch:

i $y = |f(x)|$

ii $y = f(|x|)$

iii $y = |f(|x|)|$

- 6** Repeat the steps of the previous question for the function $f(x) = |x-1| - 1$.

- 7 a** Graph $y = f(x)$, where $f(x) = \frac{1}{x-1} + 1$. Be careful to identify any intercepts with the axes and any asymptotes.

b Hence sketch:

i $y = |f(x)|$

ii $y = f(|x|)$

iii $y = |f(|x|)|$

- 8 a** Let $f(x) = x(x - 2)$.
i Sketch $y = f(x)$. **ii** Use repeated transformations to sketch $y = |f(|x|)|$.
b Repeat part **a** for the function $f(x) = (x + 1)(3 - x)$.
- 9 a** Show that $y = |2 - x|$ is neither even nor odd and graph it.
b Show that $y = |2 - |x||$ is even and use part **a** to graph this new function.
c Hence graph $y = 1 - |2 - |x||$.
d Finally graph $y = |1 - |2 - |x|||$.
- 10 a** Let $f(x)$ be any function. Explain why $g(x) = f(|x|)$ is even.
b Let $f(x)$ be an odd function. Show that $h(x) = |f(x)|$ is even.
- 11 a** When will the graphs of $y = f(x)$ and $y = |f(x)|$ be the same?
b When will the graphs of $y = f(x)$ and $y = |f(x)|$ be symmetric in the x -axis?

ENRICHMENT

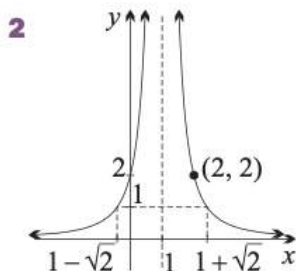
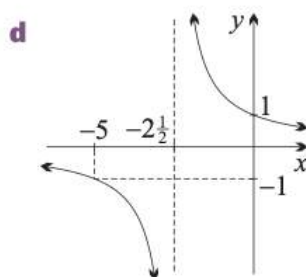
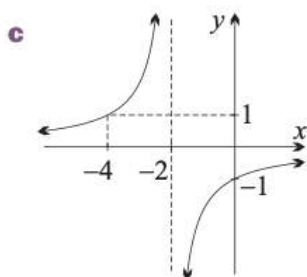
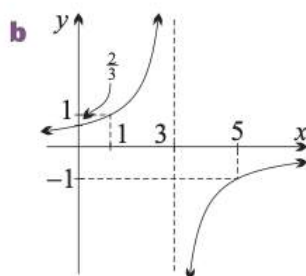
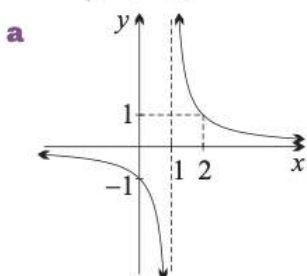
- 12 a** Read carefully the instruction in Question **1b** for graphing $y = f(|x|)$. Write a similar instruction for graphing $|y| = f(x)$.
b Test your instruction by graphing $|y| = f(x)$ for each of the functions in Question **2**.
- 13** Sketch $|y| = |x|$.
- 14 a** Describe the graph of $y = f(-|x|)$ in terms of the graph of $y = f(x)$.
b What type of symmetry must $y = f(-|x|)$ possess?
- 15** Let $f(x)$ be any function, and let $g(x) = f(|x|)$ and $h(x) = \frac{1}{2}(f(x) + f(-x))$.
a Prove that both $g(x)$ and $h(x)$ are even functions.
b Are $g(x)$ and $h(x)$ always the same function? If so then prove it, otherwise give a counter-example.
- 16 a** Investigate the graphs of the sequence of functions
$$y = |x|, \quad y = |1 - |x||, \quad y = |1 - |1 - |x|||, \quad \dots$$

b Show that the 2nd, 4th, 8th, ... functions in this sequence can be simplified to
$$y = |1 - |x||, \quad y = |1 - |2 - |x|||, \quad y = |1 - |2 - |4 - |x|||, \quad \dots$$

Exercise 6C

1 In each case $y \rightarrow 0$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

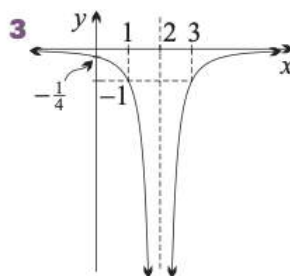
- a** i Domain: $x \neq 1$. When $x = 0$, $y = -1$.
 ii When $y = 1$, $x = 2$. When $y = -1$, $x = 0$.
 v Vertical asymptote: $x = 1$. As $x \rightarrow 1^+$, $y > 0$ so $y \rightarrow \infty$, and as $x \rightarrow 1^-$, $y < 0$ so $y \rightarrow -\infty$.
- b** i Domain: $x \neq 3$. When $x = 0$, $y = \frac{2}{3}$.
 ii When $y = 1$, $x = 1$. When $y = -1$ at $x = 5$.
 v Vertical asymptote: $x = 3$. As $x \rightarrow 3^+$, $y < 0$ so $y \rightarrow -\infty$, and as $x \rightarrow 3^-$, $y > 0$ so $y \rightarrow \infty$.
- c** i Domain: $x \neq -2$. When $x = 0$, $y = -1$.
 ii When $y = 1$, $x = -4$. When $y = -1$, $x = 0$.
 v Vertical asymptote: $x = -2$. As $x \rightarrow -2^+$, $y < 0$ so $y \rightarrow -\infty$, and as $x \rightarrow -2^-$, $y > 0$ so $y \rightarrow \infty$.
- d** i Domain: $x \neq -2\frac{1}{2}$. When $x = 0$, $y = 1$.
 ii When $y = 1$, $x = 0$. When $y = -1$, $x = -5$.
 v Vertical asymptote: $x = -2\frac{1}{2}$. As $x \rightarrow -2\frac{1}{2}^+$, $y > 0$ so $y \rightarrow \infty$, and as $x \rightarrow -2\frac{1}{2}^-$, $y < 0$ so $y \rightarrow -\infty$.



- i Domain: $x \neq 1$.
 ii When $y = 1$, $x = 1 + \sqrt{2}$ or

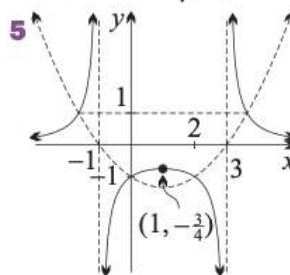
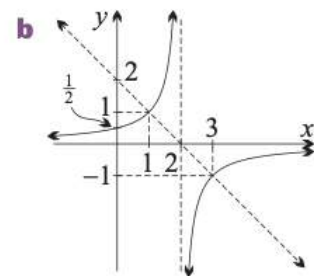
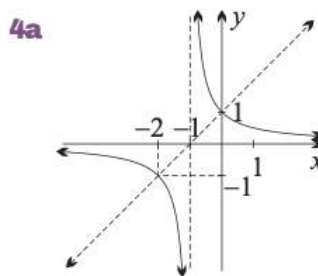
$$x = 1 - \sqrt{2}.$$

- iii H. asymptote $y = 0$, $y \rightarrow 0$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.
 v V. asymptote $x = 1$. As $x \rightarrow 1^+$, $y > 0$ so $y \rightarrow \infty$ and as $x \rightarrow 1^-$, $y > 0$ so $y \rightarrow \infty$.



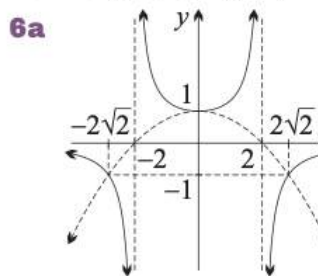
- i Domain: $x \neq 2$.
 ii When $y = -1$, $x = 1$ or 3 .

- iii H. asymptote $y = 0$, $y \rightarrow 0$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.
 v V. asymptote $x = 2$. As $x \rightarrow 2^+$, $y < 0$ so $y \rightarrow -\infty$, and as $x \rightarrow 2^-$, $y < 0$ so $y \rightarrow -\infty$.

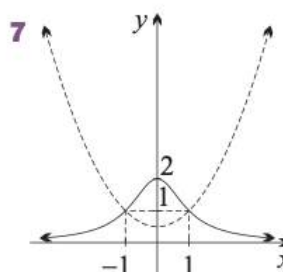


The curves also meet where $x = 1 - \sqrt{7}$ and $x = 1 + \sqrt{7}$.

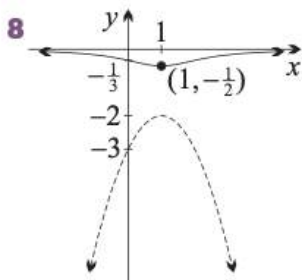
- c Range: $y \geq -\frac{4}{3}$.
 e Range: $y \leq -\frac{3}{4}$ or $y > 0$.



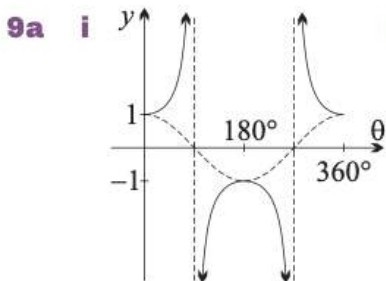
- b Range: $y \leq 1$.
 c Range: $y < 0$ or $y \geq 1$.



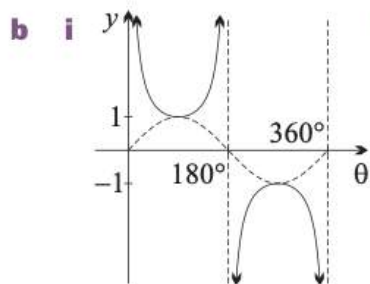
- b $\frac{1}{2}$
 d As $x \rightarrow \infty$ or $x \rightarrow -\infty$, $y \rightarrow 0$.
 e 2



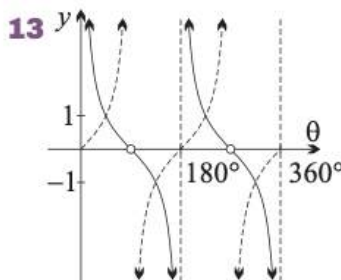
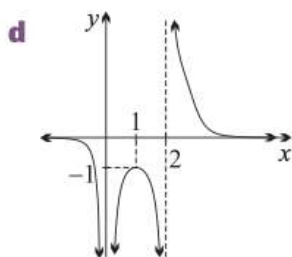
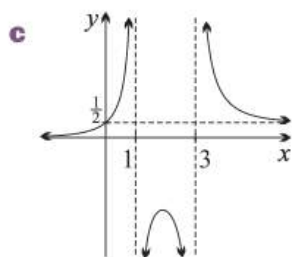
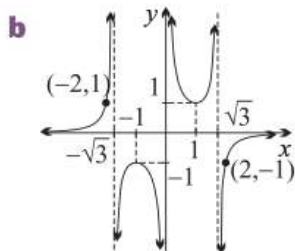
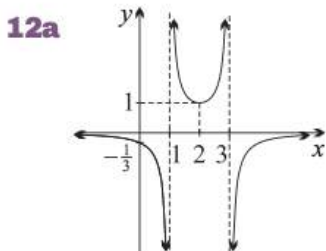
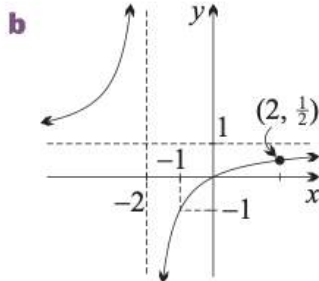
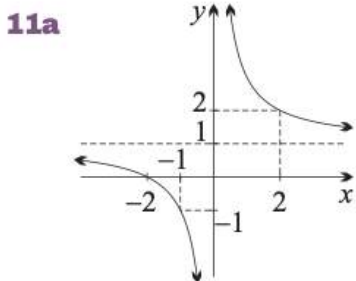
- a** -2
c As $x \rightarrow \infty$ or $x \rightarrow -\infty$, $y \rightarrow 0$.
d $-\frac{1}{2}$



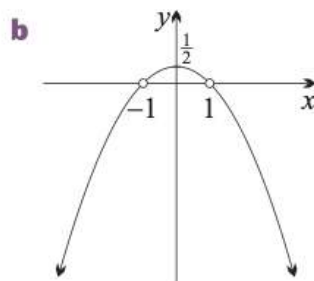
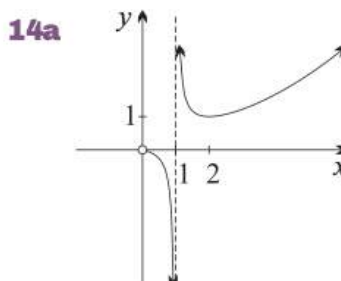
- ii** Domain:
 $0^\circ \leq \theta \leq 360^\circ$,
 except that $\theta \neq 90^\circ$
 and $\theta \neq 270^\circ$.
 Range: $y \leq -1$ or
 $y \geq 1$.



- ii** Domain:
 $0^\circ \leq \theta \leq 360^\circ$,
 except that $\theta \neq 0^\circ$,
 $\theta \neq 180^\circ$ and
 $\theta \neq 360^\circ$.
 Range: $y \leq -1$ or
 $y \geq 1$.



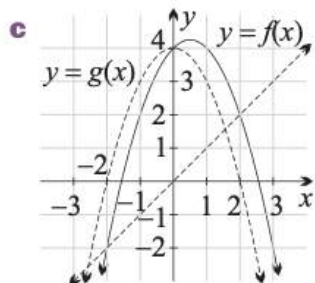
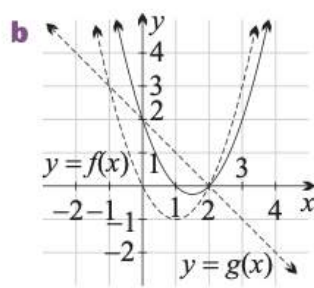
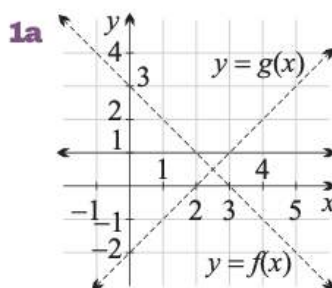
- a** Domain: $0^\circ \leq \theta \leq 360^\circ$ except that $\theta \neq 90^\circ$ and $\theta \neq 270^\circ$.
b $\tan \theta = 0$ at $\theta = 0^\circ$, $\theta = 180^\circ$ and $\theta = 360^\circ$.
c Domain: $0^\circ < \theta < 360^\circ$, except that $\theta \neq 90^\circ$, $\theta \neq 180^\circ$ and $\theta \neq 270^\circ$.
d 0
e Range: $y \neq 0$.

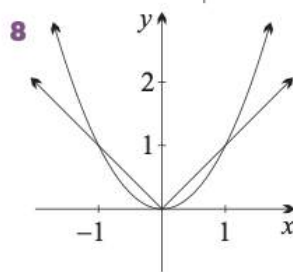
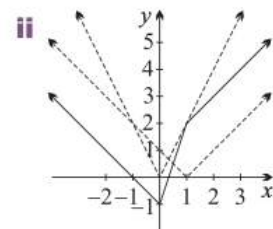
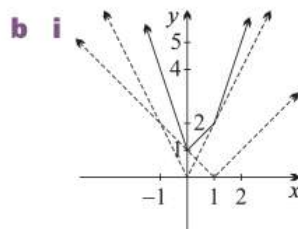
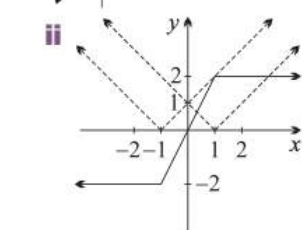
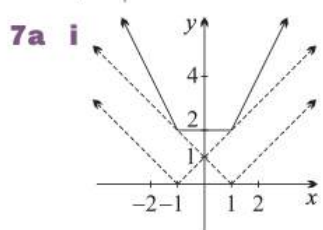
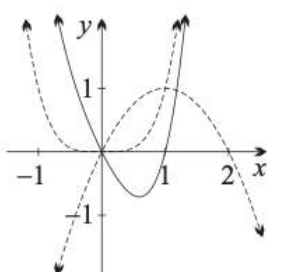
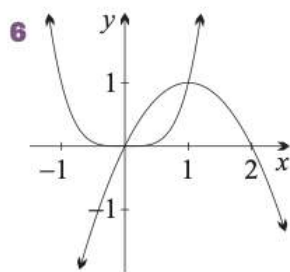
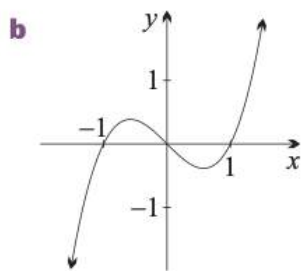
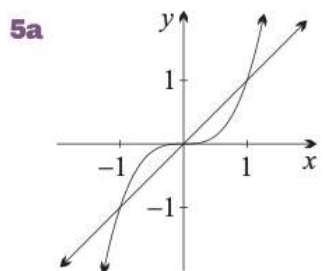
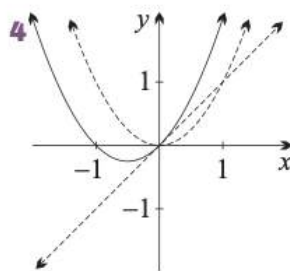
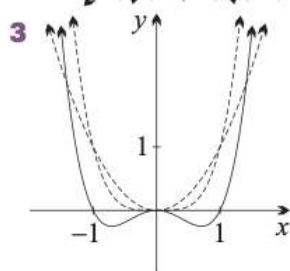
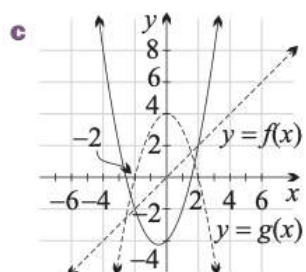
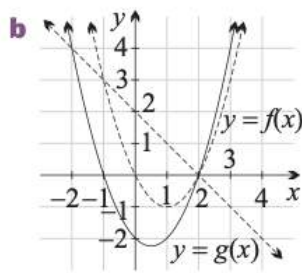
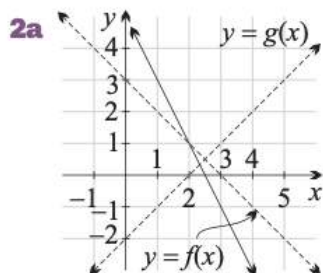


- 15** The problem is that zero does not have a reciprocal. For example, $y = -x^2$ has a maximum of 0 when $x = 0$, and $y = \frac{-1}{x^2}$ has an asymptote at $x = 0$, not a minimum. The statement should be, 'When one curve has a non-zero local maximum, the other curve has a non-zero local minimum.'

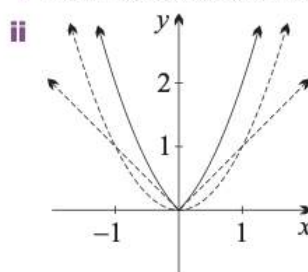
- 16** $y = x - 2$, for $x \neq 2$

Exercise 6D

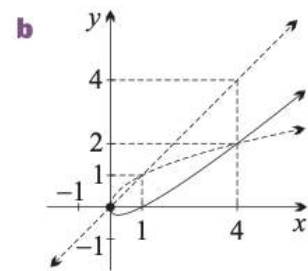
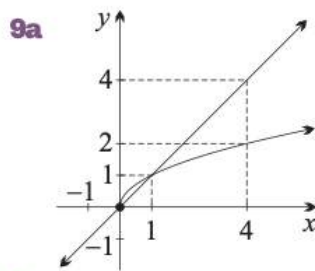
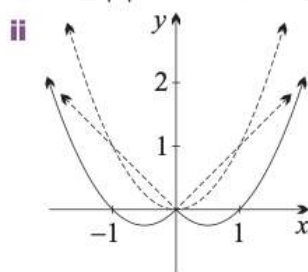




a i Both functions are always positive.



b i $x^2 \leq |x|$ whenever $-1 \leq x \leq 1$



10a $y = d(x)$ has x -intercepts at $x = -1, 2$.

b At $x = -2$, $d(-2) = 0 - g(-2) = 2$

c $d(1) = f(1) - g(1) = 3 - 1 = 2$.

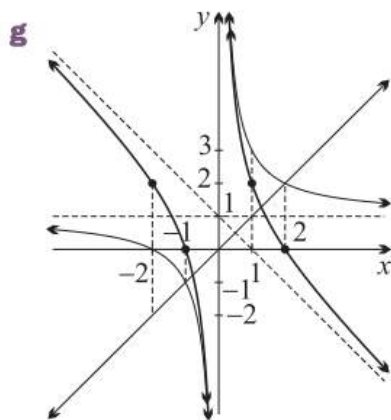
d i As $x \rightarrow 0^+$, $f(x) \rightarrow \infty$ and $g(x) \rightarrow 0$ so $d(x) \rightarrow \infty$.

ii As $x \rightarrow 0^-$, $f(x) \rightarrow -\infty$ and $g(x) \rightarrow 0$ so $d(x) \rightarrow -\infty$.

e i As $x \rightarrow +\infty$, $f(x) \rightarrow 1$ and $g(x) \rightarrow +\infty$ so $d(x) \rightarrow +\infty$

ii As $x \rightarrow -\infty$, $f(x) \rightarrow 1$ and $g(x) \rightarrow -\infty$ so $d(x) \rightarrow -\infty$

f When x is large, $f(x) \div 1$ so $d(x) \div 1 - x$.

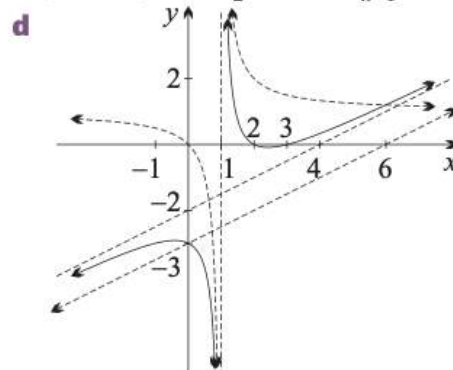


	$f(x)$ even, $g(x)$ odd
$s(x)$	neither
$d(x)$	neither

b $s(-x) = f(-x) + g(-x)$
 $= -f(x) - g(x)$
 $= -(f(x) + g(x))$
 $= -s(x)$

14a $x = 2, 3$.

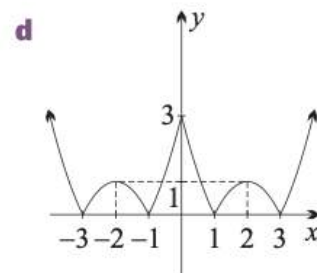
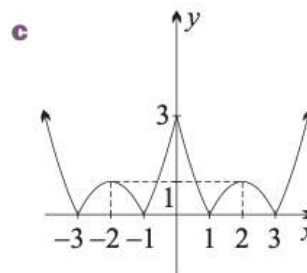
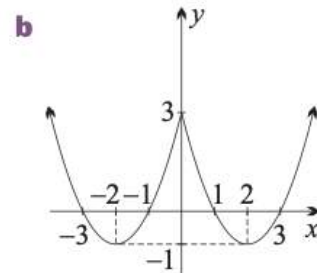
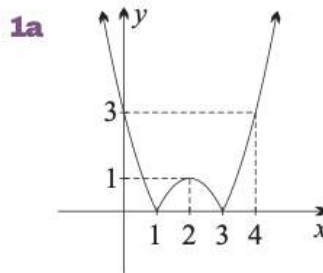
c $f(x) + g(x) - (\frac{1}{2}x - 2) = \frac{1}{x-1}$.



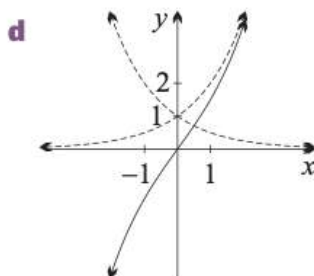
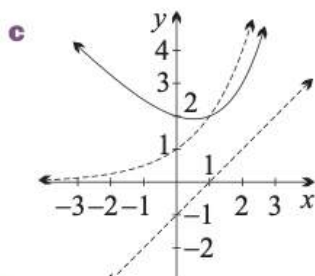
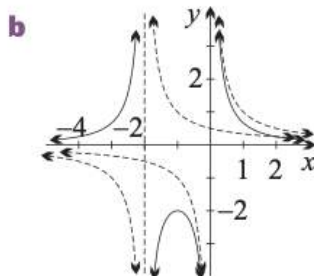
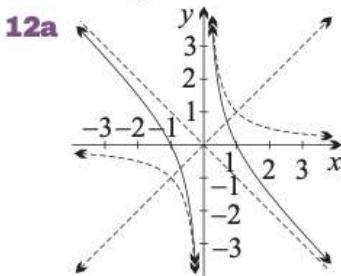
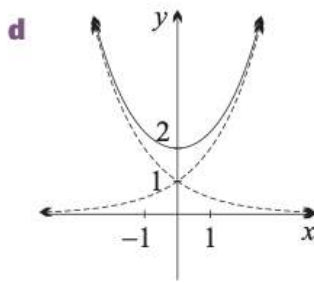
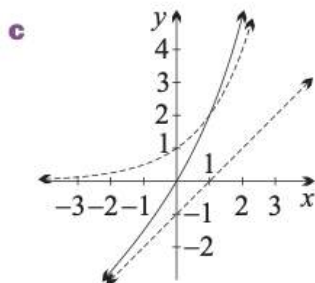
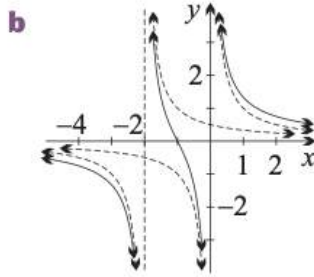
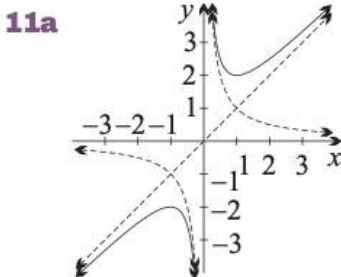
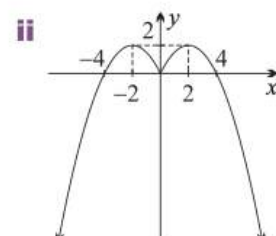
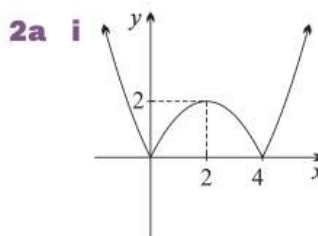
15a As $x \rightarrow \infty$ and as $x \rightarrow -\infty$,
 $s(x) - (x + 1) = \frac{1}{x} \rightarrow 0$.

b $y = 5 - x$

Exercise 6E



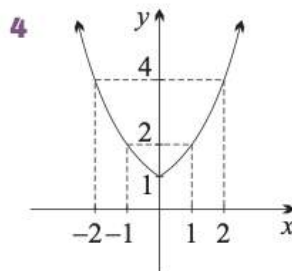
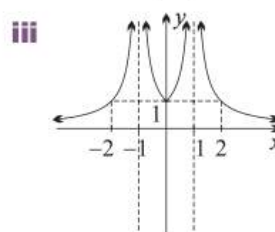
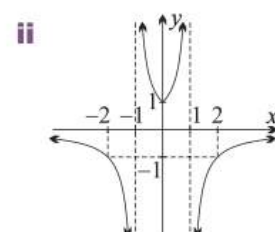
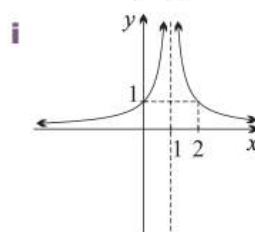
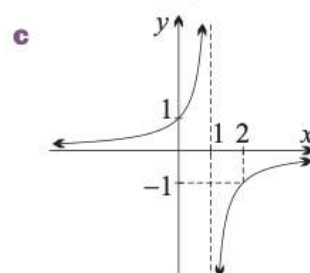
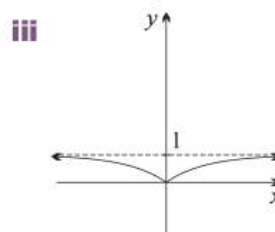
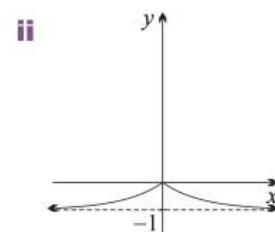
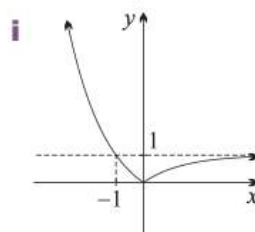
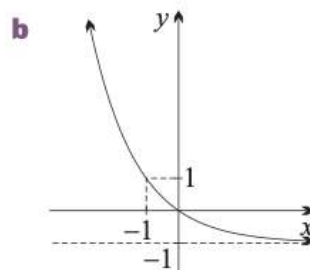
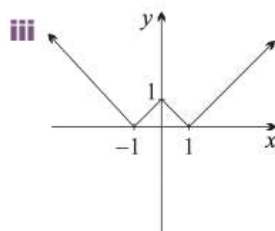
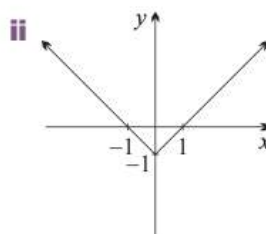
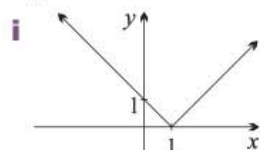
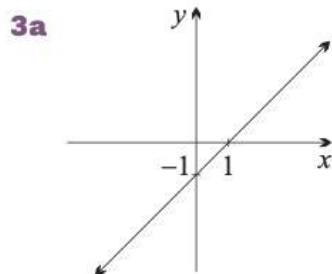
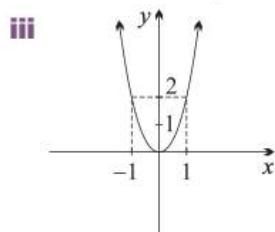
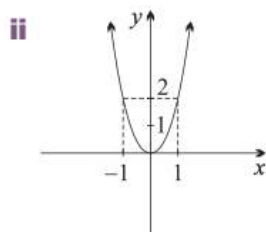
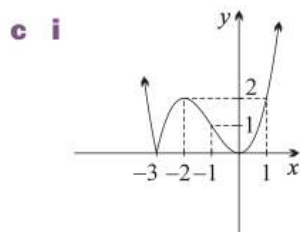
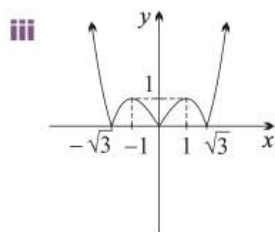
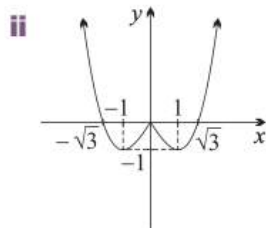
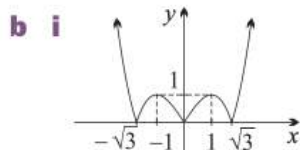
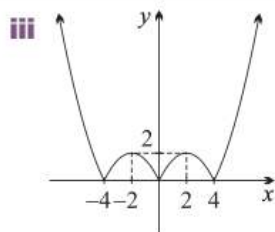
They are the same.



13a

	$f(x)$ even, $g(x)$ even
$s(x)$	even
$d(x)$	even

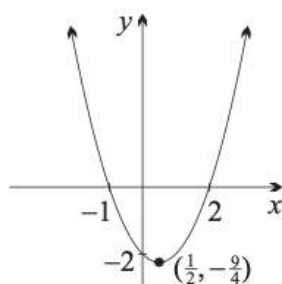
	$f(x)$ odd, $g(x)$ odd
$s(x)$	odd
$d(x)$	odd



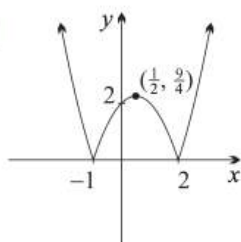
a $2^x > 0$ for all x .

b $y = \begin{cases} 2^{-x}, & x < 0 \\ 2^x, & x \geq 0 \end{cases}$

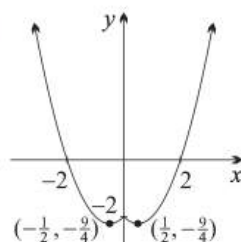
5a



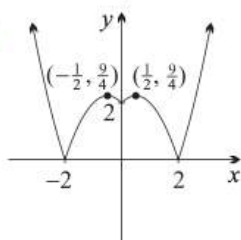
b i



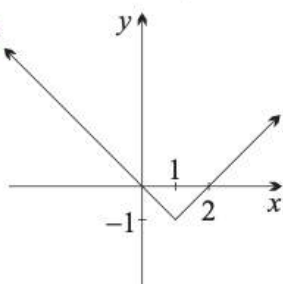
ii



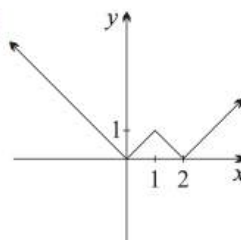
iii



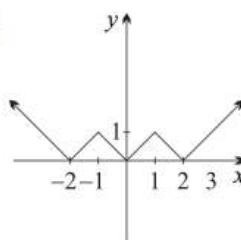
6



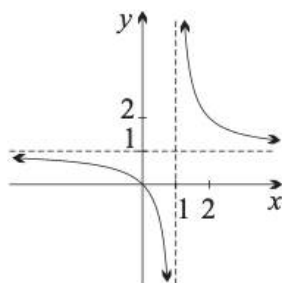
a i



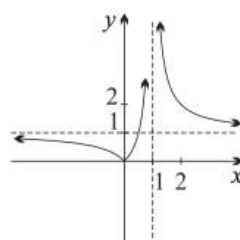
iii



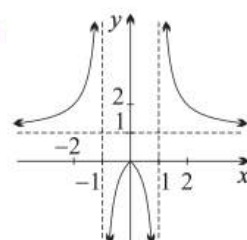
7a



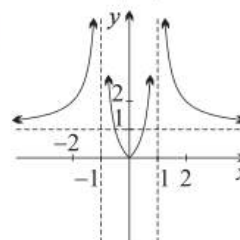
b i



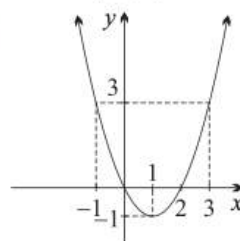
ii



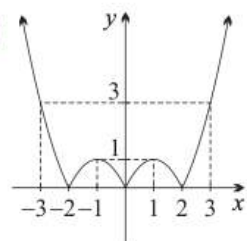
iii



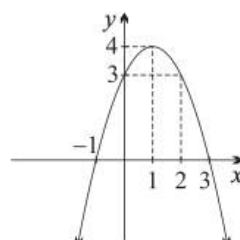
8a i



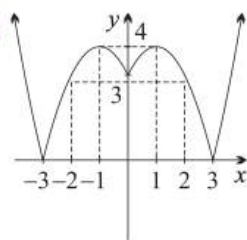
ii



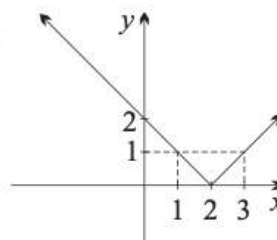
b i



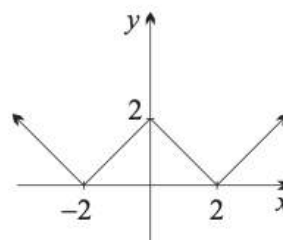
ii



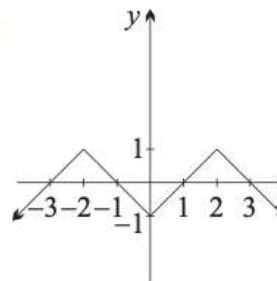
9a



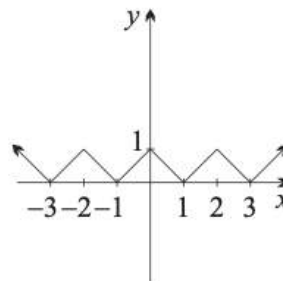
b



c



d



10a $g(-x) = f(|-x|) = f(|x|) = g(x)$

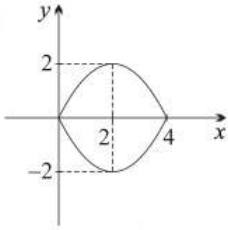
b $h(-x) = |f(-x)| = |-f(x)| = |f(x)| = h(x)$

11a When $f(x) \geq 0$ for all x .

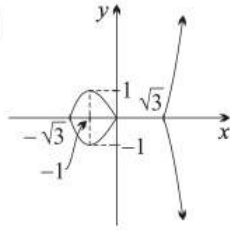
b When $f(x) \leq 0$ for all x .

12a Graph only the parts of $y = f(x)$ that are above the x -axis. Then add the reflection of these parts in the x -axis.

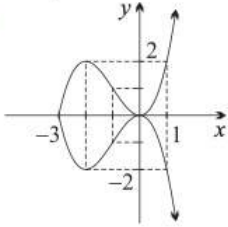
b i



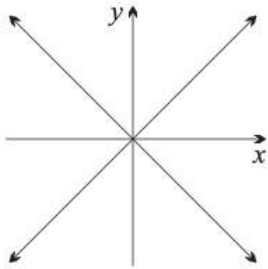
ii



iii



13



The two lines of the graph are inclined at 45° to the axes.

14a It is the parts of $y = f(x)$ that are to the left of the y -axis plus the reflection of those parts in the y -axis.

b It is even: $f(-| -x|) = f(-|x|)$

15b They need not be the same. Investigate the situation when $f(x) = x^2 - x$.